

🚀 StudyOS AI — Full Implementation Plan (Free MVP)

Project Vision

StudyOS AI is an AI-powered operating system for students that combines:

- AI Tutor
- Smart Notes
- PDF Intelligence
- Quiz Generator
- Flashcards
- Study Planner
- Analytics

The first goal is **not** to build every feature. The goal is to launch a working MVP that real students can use.

Phase 0 — Planning

Deliverables

- Project name
- Brand identity
- Logo
- Color palette
- Design system
- Folder structure
- GitHub repository
- Feature roadmap
- Database schema

Output

```
studynos-ai/  
├── frontend/  
├── backend/  
├── database/  
├── docs/  
├── assets/  
└── README.md
```

Phase 1 — UI/UX Design

Design every screen first in Google Stitch.

Don't write code.

Screens

Landing Page

Login

Signup

Forgot Password

Dashboard

AI Chat

Notes

PDF Viewer

Quiz

Flashcards

Planner

Analytics

Settings

Profile

404 Page

Loading States

Empty States

Phase 2 — Frontend

Create React App

`npm create vite@latest`

Install

Tailwind

React Router

Framer Motion

shadcn/ui

Lucide

TanStack Query

React Hook Form

Zod

Folder Structure

src/

components/

layouts/

pages/

hooks/

contexts/

services/

api/

utils/

constants/

types/

assets/

Phase 3 — Design System

Create reusable components.

Components

Button

Input

Card

Modal

Drawer

Sidebar

Navbar

Avatar

Badge

Progress

Tabs

Tooltip

Toast

Skeleton

Dialog

Dropdown

Accordion

Never duplicate UI.

Phase 4 — Authentication

Use Supabase.

Features

- Login
- Signup
- Logout
- Forgot Password
- Google Login
- Protected Routes
- Session Management

Phase 5 — Landing Page

Sections

Hero

Features

Benefits

How It Works

Testimonials

Pricing

FAQ

Footer

Deploy.

Now you already have a website.

Phase 6 — Dashboard

Create dashboard.

Widgets

Welcome

Study Streak

Calendar

Today's Goal

Assignments

Recent Chats

Recent Notes

Study Analytics

Quick Actions

Phase 7 — Database

Create tables.

profiles

notes

folders

chat_sessions

chat_messages

documents

flashcards

quizzes

quiz_attempts

tasks

study_sessions

analytics

notifications

Phase 8 — Notes

CRUD

Create

Read

Update

Delete

Search

Filter

Store in Supabase.

Phase 9 — AI Chat

Backend

FastAPI

Gemini

Features

Chat

History

Streaming

Markdown

Code Highlight

Math

Copy

Regenerate

Suggested Questions

Phase 10 — File Upload

Upload

PDF

DOCX

PPTX

TXT

Store in Supabase Storage.

Phase 11 — PDF Processing

Backend

Read PDF

Extract Text

Chunk Text

Send to Gemini

Generate Summary

Generate Notes

Generate Quiz

Generate Flashcards

Phase 12 — Quiz Generator

Input

PDF

Notes

Topic

Output

MCQ

Short Answer

Coding Questions

Difficulty

Timer

Save results.

Phase 13 — Flashcards

Generate

Question

Answer

Hint

Difficulty

Add

Review

Progress

Spaced Repetition

Phase 14 — Planner

Features

Calendar

Tasks

Pomodoro

Goals

Daily Schedule

Weekly Schedule

AI Study Plan

Phase 15 — Analytics

Charts

Study Hours

Quiz Score

Weak Topics

Strong Topics

AI Usage

Learning Trend

Study Heatmap

Weekly Report

Phase 16 — Settings

Profile

Theme

Notifications

Connected Accounts

Language

Delete Account

Phase 17 — Backend

FastAPI

Modules

Auth

Notes

Chat

PDF

Quiz

Planner

Analytics

Phase 18 — AI Layer

Gemini API

Functions

Explain

Summarize

Generate Notes

Generate Quiz

Generate Flashcards

Study Plan

Translate

Rewrite

Phase 19 — Storage

Supabase Storage

Folders

pdf/

notes/

avatars/

exports/

Phase 20 — Deployment

Frontend

Vercel

Backend

Render

Database

Supabase

Final Folder Structure

studynos-ai/

frontend/

src/

components/

pages/

layouts/

hooks/

contexts/

services/

api/

assets/

backend/

app/

routers/

models/

schemas/

services/

utils/

database/

docs/

README.md

API Endpoints

Auth

POST /signup

POST /login

POST /logout

GET /me

Notes

GET /notes

POST /notes

PATCH /notes/:id

DELETE /notes/:id

AI

POST /chat

POST /summarize

POST /generate-quiz

POST /generate-flashcards

POST /study-plan

PDF

POST /upload

GET /pdf/:id

DELETE /pdf/:id

Quiz

GET /quiz

POST /quiz

POST /submit

GitHub Workflow

main

develop

feature/auth

feature/dashboard

feature/chat

feature/notes

feature/pdf

feature/planner

feature/quiz

Development Order

✓ UI Design

↓

Landing Page

↓

Authentication

↓

Dashboard

↓

Notes

↓

AI Chat

↓

PDF Upload

↓

Summarizer

↓

Quiz

↓

Flashcards

↓

Planner

↓

Analytics

↓

Deployment

↓

Beta Testing

↓

Launch

MVP Checklist

- ☐ Landing Page
 - ☐ Authentication
 - ☐ Dashboard
 - ☐ AI Chat
 - ☐ Notes
 - ☐ PDF Upload
 - ☐ AI Summary
 - ☐ Quiz Generator
 - ☐ Flashcards
 - ☐ Planner
 - ☐ Analytics
 - ☐ Mobile Responsive
 - ☐ Deployment
-

After MVP (Future Versions)

Version 2

- Voice conversations
- OCR for handwritten notes
- AI mind maps
- Browser extension

Version 3

- Mobile app (React Native)
- Offline mode
- Real-time collaboration
- Group study rooms

Version 4

- AI career coach
- Resume builder
- Interview practice
- Internship recommendations

How we'll build it together

Instead of trying to finish everything in one go, we'll treat this like a real startup.

- **Sprint 1:** Landing page + Design System
- **Sprint 2:** Authentication + Dashboard
- **Sprint 3:** AI Chat (ChatGPT-style)
- **Sprint 4:** Notes + PDF Intelligence
- **Sprint 5:** Quiz + Flashcards + Planner
- **Sprint 6:** Analytics + Deployment + Launch

This keeps the project manageable, gives you something usable after every sprint, and produces a strong portfolio piece while staying completely within the free tiers of the tools you've chosen.